

Deep Learning for Natural Language Processing

Assignment- 4

TYPE OF QUESTION: MCQ

Number of questions: 7 Total marks: $4 * 1 + 3 * 2$ [Q5, Q6, Q7 carries two marks each]

Question 1: Which statement comparing greedy decoding and beam search is TRUE?

1. Greedy decoding is the special case of beam search with beam width $k = 1$, and both are deterministic.
2. Beam search is a stochastic decoding strategy, whereas greedy decoding is deterministic.
3. Beam search with any $k > 1$ is guaranteed to return the globally optimal sequence.
4. Greedy decoding maximises the total sequence log-probability by construction.

Answer: 1

Greedy decoding keeps exactly one hypothesis at each step, so it is beam search with $k = 1$. Both produce a deterministic output (unlike sampling methods). Beam search explores more of the search space than greedy but gives no optimality guarantee, and greedy's locally best choices can be globally suboptimal.

Question 2: A named-entity recognition system must recognise 7 distinct entity types and uses the BIO tagging scheme. How many distinct tags does the per-token classifier have to choose from?

1. 7
2. 14
3. 15
4. 16

Answer: 3

BIO gives n B-tags, n I-tags and a single O-tag, i.e. $2n + 1$. For $n = 7$: $2(7) + 1 = 15$.

Question 3: An RNN language model is trained on a 4-token sequence. At the four timesteps it assigns the following probabilities to the correct next token: 0.5, 0.25, 0.125, 0.0625. Using logarithms to base 2, what is the average cross-entropy loss per token?

1. 0.625
2. 1.25
3. 2.5
4. 10

Answer: 3

The per-timestep loss is $L_{CE} = -\log y_{gold}$, and the training objective is $(1/T) \sum_t L_{CE}$.

$$-\log_2 0.5 = 1$$

$$-\log_2 0.25 = 2$$

$$-\log_2 0.125 = 3$$

$$-\log_2 0.0625 = 4$$

Sum = 10; average = $10 / 4 = 2.5$ bits per token.

Question 4: A sequence-to-sequence model with dot-product attention translates a source sentence of 12 tokens into a target sentence of 9 tokens. How many attention weights α_{ij} are computed in total over the whole decoding process?

1. 12
2. 21
3. 9
4. 108

Answer: 4

At each decoder step the model computes one weight per encoder state, i.e. 12 weights. Over 9 decoder steps: $9 \times 12 = 108$.

Question 5: A Bi-LSTM is used for NER. Word embeddings are 100-dimensional; the forward and backward LSTMs each have a hidden state of 64 dimensions. There are 4 entity types and the BIO scheme is used. The output layer is applied to the concatenated hidden state. How many parameters must be trained, excluding biases and excluding the embedding table?

1. 85120
2. 43136
3. 84544
4. 83968

Answer: 1

Number of tags = $2(4) + 1 = 9$. Each LSTM direction has 4 blocks (forget gate, input gate, output gate, cell candidate), each with a W (64×100) and a U (64×64)

Per direction: $4 \times (64 \times 100 + 64 \times 64) = 4 \times (6,400 + 4,096) = 4 \times 10,496 = 41,984$

Two directions: $2 \times 41,984 = 83,968$

Output layer on the concatenated state ($64 + 64 = 128$): $9 \times 128 = 1,152$

Total = $83,968 + 1,152 = 85,120$

Question 6: A sequence-to-sequence RNN with dot-product attention has 3 encoder hidden states (2-dimensional):

$h_1 = [2, 0]$, $h_2 = [0, 2]$, $h_3 = [1, 1]$

and the relevant decoder hidden state is $h^d_{i-1} = [1, -1]$. Which is the resulting context vector c_i (to 2 decimal places)?

1. $[1.85, 0.15]$
2. $[1.00, 0.33]$
3. $[0.15, 1.85]$
4. $[1.50, 0.50]$

Answer: 1

Scores: $h^d \cdot h_1 = 2$, $h^d \cdot h_2 = -2$, $h^d \cdot h_3 = 0$.

Softmax over $[2, -2, 0]$: $e^2 \approx 7.389$, $e^{-2} \approx 0.135$, $e^0 = 1$; $\text{sum} \approx 8.524 \rightarrow \alpha \approx [0.867, 0.016, 0.117]$

Context vector $c_i = \sum_j \alpha_{ij} h_j = 0.867 \cdot [2, 0] + 0.016 \cdot [0, 2] + 0.117 \cdot [1, 1] = [1.734 + 0.117, 0.032 + 0.117] \approx [1.85, 0.15]$

Question 7: A machine-translation system uses an RNN encoder–decoder with dot-product attention. The source vocabulary is 30,000, the target vocabulary is 20,000, both sides use 200-dimensional embeddings, and the encoder and decoder hidden states are both 150-dimensional. Counting the two embedding tables, both recurrent units, and the output projection to the target vocabulary (and ignoring biases), how many parameters are there in total?

1. 9,105,000
2. 13,105,000

3. 13,405,000
4. 10,105,000

Answer: 2

Source embeddings: $30,000 \times 200 = 6,000,000$

Encoder W (150×200) = 30,000; encoder U (150×150) = 22,500

Target embeddings: $20,000 \times 200 = 4,000,000$

Decoder W (150×200) = 30,000; decoder U (150×150) = 22,500

Output projection V: $20,000 \times 150 = 3,000,000$

Dot-product attention: 0 additional parameters

Total = $6,000,000 + 30,000 + 22,500 + 4,000,000 + 30,000 + 22,500 + 3,000,000 = 13,105,000$
